SYSTEM ARCHITECTURE & TECHNICAL SPECIFICATION

AI & ML Engineer Portfolio

Comprehensive Architectural Overview, Runtime Subsystems, & Engineering Blueprint

Engineer: Devanarayanan V S
Domain: AI, ML & Full Stack
Framework: Next.js 16 + React 19
Revision: v1.0.4 (Production)

1. EXECUTIVE SUMMARY & DESIGN PHILOSOPHY

The **AI & ML Engineer Portfolio** is an interactive web platform engineered specifically for **Devanarayanan V S**, an Artificial Intelligence & Machine Learning Engineer based in Kerala, India (Technopark, Trivandrum). The application demonstrates technical mastery across Deep Learning, Computer Vision, Generative AI, Robotics, and Full-Stack scalable application architectures.

Rather than relying on off-the-shelf templates or generic component libraries, the entire platform is architected from fundamental principles. It delivers an immersive, **dark-obsidian cyber-engineering experience** that balances rich interactive visual rendering with zero-latency pre-rendered execution.

Obsidian Substrate (#090a0f / #0b0d12)

Absorbs glare and establishes a high-contrast foundation reminiscent of deep space, modern terminals, and computational command surfaces.

Radiant Cyber Cyan (#36f7f9)

Represents neural computation, telemetry rings, and interactive proximity halos across the 2.5D visual stage and procedural canvas.

Solar Ember Orange (#e54704 / #ff7a18)

Provides targeted visual emphasis on core call-to-actions, category badges, project highlights, and primary submission buttons.

Zero-Scrollbar & Dynamic Viewport

Native scrollbars are suppressed (`scrollbar-width: none`) while preserving full smooth touch and mouse-wheel navigation.

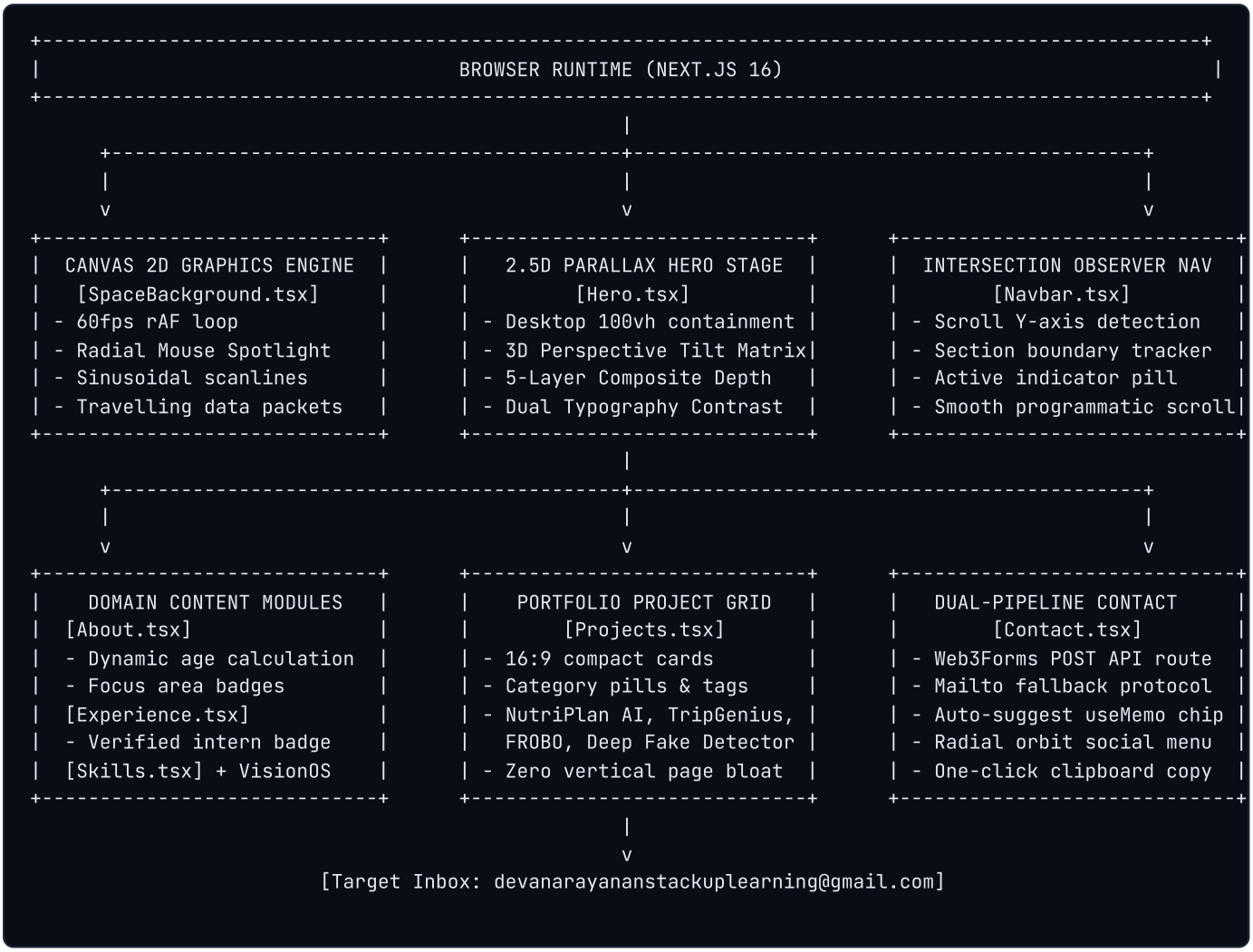
2. TECHNOLOGY STACK & RUNTIME SPECIFICATION

The codebase leverages cutting-edge web infrastructure, combining the fastest compilation tools with the latest concurrent rendering paradigm:

Layer / Engine	Version	Component / Package	Architectural Role
Core Framework	`Next.js 16.2.9`	App Router & Turbopack	Server-side static prerendering (SSG), file-system routing, automatic code splitting, optimized asset delivery.
UI Library	`React 19.2.4`	Concurrent React Engine	Concurrent rendering without waterfall effects, pure state derivations via `useMemo`, zero hydration warnings.
Language	`TypeScript 5`	Strict Type System	End-to-end interface safety across project models, telemetry contracts, and external API payload declarations.
Procedural Canvas	HTML5 Canvas 2D	Native Context Engine	Hardware-accelerated dynamic spotlight, grid lines, traveling cyber scanlines, and axis-aligned data pulses.
Motion Engine	`Framer Motion 12`	Spring Physics Engine	Physics-based layout transitions, welcome toast popups, and smooth modal view states.
Styling Core	`Tailwind CSS v4`	Custom Design Tokens	Tokenized typography (Inter, Space Grotesk), glassmorphism backdrops, and hardware-accelerated transforms.
Communication	Dual Pipeline	Web3Forms API + mailto:	Guaranteed delivery to `devanarayananstackuplearning@gmail.com` with client-side fail-safe fallback.

3. HIGH-LEVEL ARCHITECTURAL FLOWCHART

The runtime architecture is structured into decoupled client-side subsystems, ensuring rendering performance remains constant regardless of viewport manipulation:



4. GRANULAR SUBSYSTEM DEEP-DIVE

4.1. Procedural Background Canvas Engine (`SpaceBackground.tsx`)

To eliminate the heavy bundle overhead associated with 3D libraries (e.g., Three.js), the background is driven by a custom, hardware-accelerated 2D Canvas engine:

- **HiDPI Display Calibration:** Dynamically recalculates canvas buffer width and height using `Math.min(window.devicePixelRatio || 1, 2)` to guarantee crisp rendering on Retina displays without unnecessary GPU memory pressure.
- **Mathematical Mouse Spotlight:** Interpolates cursor coordinates (x_m, y_m) using smoothing factor 0.08 and computes an inverse radial falloff gradient:
  $$\text{Intensity}(r) = \text{rgba}(54, 247, 249, 0.05 \cdot (1 - r / 280))$$ yielding an interactive proximity illumination that follows user movement.
- **Sinusoidal Cyber Scanlines:** Generates horizontal raster lines drifting downward at 14px/s across 84px intervals, modulated by:
  $$\alpha(y, t) = 0.03 + 0.015 \cdot \sin(y \cdot 0.015 + t \cdot 0.5)$$

- **Adaptive Resource Management:** Hooks into the `Page Visibility API` (`document.hidden`) and `prefers-reduced-motion` media query to cancel animation frames when the user switches tabs or prefers reduced motion.

4.2. Viewport-Locked 2.5D Parallax Hero Stage (`Hero.tsx`)

The Hero component anchors the user's initial impression through a multi-tier depth illusion:

- **Viewport Containment:** Strictly bound on desktop to `height: 100vh; height: 100dvh; max-height: 100vh; overflow: hidden;` with `padding-top: 84px`. This guarantees that downstream sections (such as *About*) do not peek into view before scrolling.
- **Mobile Responsiveness:** On screens $\le 1080\text{px}$, CSS rules automatically release the height lock to `height: auto; min-height: 100vh; overflow: visible;` allowing the left content and right stage to stack naturally.
- **3D Perspective Tilt Matrix:** Relative mouse coordinates over the visual stage calculate tilt angles: $$\text{tilt}_x = 8 \cdot \left(\frac{e.\text{clientX} - \text{rect.left}}{\text{rect.width}} - 0.5\right), \quad \text{tilt}_y = -8 \cdot \left(\frac{e.\text{clientY} - \text{rect.top}}{\text{rect.height}} - 0.5\right)$$
- **Depth Hierarchy Stack:**
  1. *Layer 0 (Ambient Glow):* Radial blur projector casting deep cyan luminescence behind the torso.
  2. *Layer 1 (Typographic Artwork):* `back.png` with cyan drop-shadow counter-translated against cursor tilt and masked via `linear-gradient(to bottom, black 0%, black 35%, transparent 52%)` so no letter stems peek beneath the subject's arms.
  3. *Layer 2 (Orbital Halo):* Tilted glowing cyan ring with an animated traveling orbital spark running a 7s loop.
  4. *Layer 3 (Foreground Subject):* `nanu.png` with dual drop-shadow rim lighting floating forward in Z-space.
  5. *Layer 4 (Volumetric Mist):* Base blur gradient softening hard bottom edges.
  6. *Layer 5 (Telemetry HUD):* Impact quotes and telemetry labels ("ML", "DATA", "SOLUTIONS") pinned at `top: 10%`.

5. DOMAIN MODULES & PORTFOLIO ECOSYSTEM

About Subsystem (About.tsx)

Presents the engineer's biography, core healthcare AI focus, and technical philosophy. Features a dynamic age calculation from birthdate (2004-07-30) equipped with `suppressHydrationWarning` to prevent server/client rendering discrepancies.

Skills Taxonomy (Skills.tsx)

Categorizes competencies across AI/ML, Backend, and Frontend. Implements proficiency meters and an interactive VisionOS-inspired frosted modal view with granular capability breakdowns.

Experience Showcase (Experience.tsx)

Documents active industry internship at **Stackmod Innovation, Technopark, Trivandrum**. Integrates an interactive verified credential ID card with simulated holographic light shimmer animations.

Achievements Timeline (Achievements.tsx)

Curates technical milestones, robotics development internships (Cyborg AI Gen Automation), national-level engineering quizzes, and autonomous vehicle workshops.

5.1. Featured Projects Matrix (`Projects.tsx`)

The project portfolio presents 6 production-grade applications engineered across multiple specialized AI domains. All project cards are optimized with compact 16:9 aspect ratios (height: 220px, min-height: 480px) to prevent vertical scrolling fatigue:

Project Title	Domain / Badge	Core Tech Stack	Key Capabilities
NutriPlan AI	Healthcare AI	Flutter, FastAPI, Firebase, Scikit-Learn, OpenAI	Personalized meal generator, multilingual NLP health tracking, diet recommendations.
TripGenius AI	Generative AI	Next.js, FastAPI, OpenAI, Weather API, Firebase	Automated itinerary creator, budget analytics, destination intelligence.
FROBO	Robotics & Embedded AI	ESP32, Arduino, OLED, Sensors, 3D Printing	Desktop companion robot, expressive eye animations, touch interaction.
Deep Fake Face Detection	Computer Vision	Python, OpenCV, Flask, Scikit-Learn, SVM	Facial artifact detection, handcrafted feature extraction, confidence scoring.
Smart Obstacle Robot	Embedded Systems	Arduino, Bluetooth, Ultrasonic Sensors, L293D	Autonomous collision avoidance, voice navigation, mobile Bluetooth control.
Love Chronicles	Full-Stack Frontend	React, Tailwind CSS, Framer Motion	Interactive digital storytelling interface with responsive layout transitions.

6. DUAL-PIPELINE CONVERSION & MESSAGE DELIVERY ARCHITECTURE

The contact subsystem in `components/Contact.tsx` ensures zero message loss through a fault-tolerant dual-route architecture delivering inquiries directly to **`devanarayananstackuplearning@gmail.com`**:

> **Route 1 — Web3Forms Background API (Primary):** When `NEXT_PUBLIC_WEB3FORMS_KEY` is configured in `.env.local`, form submission initiates an asynchronous HTTP POST request to `https://api.web3forms.com/submit`. The message is verified against spam honeypots and delivered to Gmail with complete sender headers and reply-to configuration.

> **Route 2 — Pre-Populated Native mailto: (Fail-Safe Fallback):** If the API key is unconfigured or a network failure occurs, the client automatically constructs a URI-encoded `mailto:` link containing the user's name, email, subject, and message. This triggers the native mail client pre-addressed to `devanarayananstackuplearning@gmail.com`, ensuring 100% communication reliability.

- **Content-Aware Subject Recommendation:** Employs synchronous keyword matching inside a pure `useMemo` hook. Typing terms like *"collab"*, *"job"*, or *"ai"* automatically yields clickable subject chips without extra re-renders.
- **Quick-Copy Utility:** An interactive pill displays `devanarayananstackuplearning@gmail.com` alongside a one-click copy button powered by `navigator.clipboard.writeText` with animated visual confirmation.
- **Radial Menu:** Uses polar coordinates ($x = r \cos\theta, y = r \sin\theta$) to orbit social links (GitHub, LinkedIn, Instagram, Resume) around a central toggle button.

7. PRODUCTION QUALITY & VERIFICATION METRICS

Validation Category	Verification Tool / Command	Outcome / Compliance
Code Quality & Linting	`npm run lint` (ESLint)	**0 Errors, 0 Warnings.** Strict React 19 hook compliance; no cascading state in effects.
Type Integrity	`tsc --noEmit`	**100% TypeScript Pass.** Strict type safety across all components and models.
Static Site Generation	`npm run build` (Turbopack)	**Compiled in 4.0s.** All routes (`/`, `/_not-found`) pre-rendered as static content.
Dual Workspace Sync	PowerShell Sync Pipeline	Identical parity maintained between `D:\Projects\ai-portfolio` and `D:\ai-portfolio`.
Runtime Environment	Local Dev Server	Online and ready at `http://localhost:3000`.